

VINITH BONILA

AI ENGINEER | DATA SCIENTIST

Python | LLMs | RAG | NLP | Machine Learning | FastAPI | SQL
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PROFESSIONAL SUMMARY

Python developer who ships production-grade AI systems end to end. Deployed two live LLM applications backed by FastAPI services, Docker packaging, automated tests and CI, plus an automated analytics platform with 241 backend tests. Petroleum Engineering graduate carrying upstream oil and gas domain knowledge from an ONGC field internship. Seeking early-career AI engineering, machine learning and data science roles.

TECHNICAL SKILLS

Programming: Python, SQL, C

AI & NLP: RAG, LLM APIs, prompt design, embeddings, BM25, cross-encoder reranking, Reciprocal Rank Fusion, spaCy, Hugging Face, sentence-transformers, intent classification, NER, slot extraction

Machine Learning: scikit-learn, TensorFlow/Keras, supervised regression and classification, LSTM sequence models, model evaluation and benchmarking

Data Science: pandas, NumPy, exploratory data analysis, feature engineering, statistics, SQL (MySQL, PostgreSQL, SQLite, DuckDB), FAISS vector database

Backend & MLOps: FastAPI, REST API design, OOP, unit testing, Docker, Git/GitHub, GitHub Actions CI, Streamlit, Render deployment

Data & BI: Power BI, DAX, Tableau, Excel

Domain & Simulation: Upstream petroleum operations, SCADA/field data, reservoir simulation (CMG GEM, WinProp)

PROJECTS

DocMind - AI Document Assistant (RAG) | Python, FastAPI, Streamlit, FAISS, Groq LLM, Docker | [Live](#) **2026**

- Built and deployed a production-grade RAG assistant answering questions over any uploaded PDF/DOCX/TXT with grounded, page-cited, streaming responses; served via FastAPI, containerized with Docker and hardened with 24 automated tests and GitHub Actions CI.
- Engineered hybrid dense (FAISS) and sparse (BM25) retrieval fused via Reciprocal Rank Fusion with cross-encoder reranking - recall@5 of 1.00 and MRR of 0.96; an evaluation harness measured faithfulness (0.78) and answer relevance (0.82) using LLM-as-judge.

VINI AI - NLP-First Voice Assistant | Python, Streamlit, spaCy, Whisper, SQLite, Groq API | [Live](#) **2026**

- Built and deployed an NLP-first voice assistant resolving intent through semantic similarity - sentence-transformer embeddings with TF-IDF and fuzzy fallbacks - across a 19-intent catalog at 15-20 ms classification latency, replacing brittle keyword matching.
- Engineered a modular NLU pipeline (spaCy NER with intent-aware slot extraction, dependency parsing, sentiment and language detection) with live web-search RAG on Groq Llama-3.3-70B, a Whisper speech-to-text and TTS loop, SQLite persistence and a 15+ skill handler router.

AutoBI - Automated Business Intelligence Platform | Python, FastAPI, Pandas, DuckDB, PostgreSQL, Next.js | [Live](#) **2026**

- Built an analytics platform that converts raw CSV/TSV uploads into interactive dashboards, automating data profiling, cleaning, KPI discovery and chart recommendation through a FastAPI and Pandas/DuckDB pipeline.
- Shipped dashboards with dynamic filters, daily-to-yearly time aggregation, saved views, shareable links and PDF/Excel/CSV export, backed by PostgreSQL persistence and 241 automated backend tests; deployed on Render.

Data Analytics - Zomato & Namma Yatri Insights | SQL, Python, Power BI, Excel **2026**

- Traced a 2,161-search ride booking funnel across a 5-table schema, isolating the fare-estimate stage as the largest drop-off at 403 lost searches - three times the combined loss from driver and customer cancellations.
- Wrote 45 SQL analysis queries using partitioned RANK and DENSE_RANK window functions across both datasets and built Power BI dashboards with DAX measures and drill-through filters to surface demand and rating patterns.

EXPERIENCE

Intern - Oil and Natural Gas Corporation (ONGC), Ankleshwar Asset **May - Jul 2025**

- Analysed SCADA-based real-time field monitoring data across gas gathering, three-stage compression, dehydration and metering, tracking operational parameters across upstream production facilities.
- Worked with technical teams on structured problem-solving for production and artificial lift performance reviews, translating field observations into operational efficiency findings.

LEADERSHIP & ACHIEVEMENTS

President - ShutrX Photography Club, IIPe (Jul 2025 - May 2026)

- Led a 25+ member team through a full rebrand, growing fest participation 156% (90 to 230+) and digital reach to 60K+, establishing the club as the institute's official media body.
- Owned a Rs. 3L annual budget at ~93% utilisation with zero overruns, running GeM portal procurement, vendor coordination and 100% audit-ready documentation.
- Planned event calendars and delegated work across sub-teams, reporting outcomes to faculty and institute administration.

Public Relations Officer - SPE Student Chapter, IIPe (Jul 2024 - Mar 2025): coordinated outreach, speaker communication and event promotion across stakeholder groups.

EDUCATION

B.Tech, Petroleum Engineering - Indian Institute of Petroleum and Energy (IIPe), Visakhapatnam | CGPA: 6.29 **2026**

Class XII (CBSE), PCM - Narayana College | 80.5% **2021**

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification - Stanford Online / DeepLearning.AI
- SQL Gold Badge - HackerRank
- Data Analytics Job Simulation - Quantum (Forage) | Power BI Job Simulation - PwC (Forage)
- Data Business Analytics Specialization - Jobaaj | Excel Skills for Business - Macquarie University